FA report of Flicker issue

JUNGLIM_P215HAN01.1

Date：2026/03/23

AUO

Problem Statement

● General Information

Brand Customer	JungLim	Defect Phenomenon	Flicker
SI Customer	Kortek / Samsung Medison	Defect Rate or Q'ty	3
AUO Model	P215HAN01.1	Defect Station	Samsung Medison OQC
AUO S/N	Q55A25P5KLZZZZ1900 Q55A4736QLZZZZ1900 Q55A40P8XHZZZZ1900	Chip ID	A25P5KL A4736QL A40P8XH
Array/Cell/Beol/Module	L6A/L6A/L6A/Z19	RMA No.	702510106891
Week Code	25/32	Product Provider	

● Defect Panel S/N Picture

Costomer Data

A total of 4 cases were reported in November

1. Model: P215HAN01.1
2. Customer: Kortek / Samsung Medison
3. Application: ultrasound
4. Process: Samsung Medison OQC
5. Defect: Flicker

- After aging, screen flicker occurs on the gray pattern during On/Off switching.

6. Qty: 3 unit

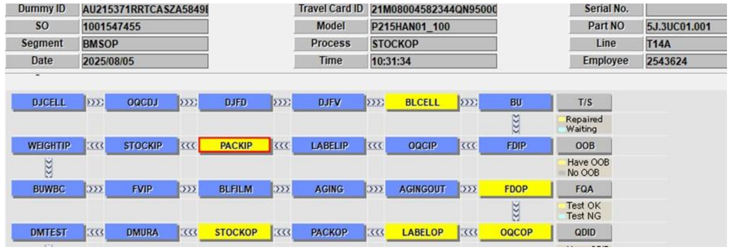
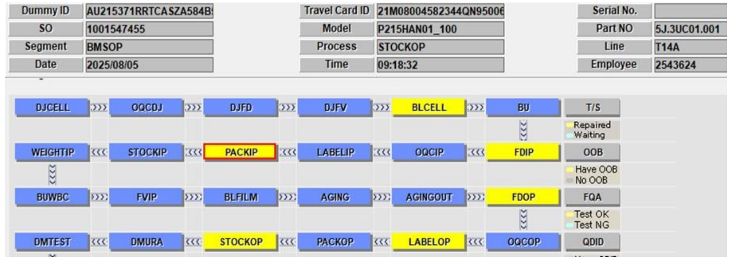
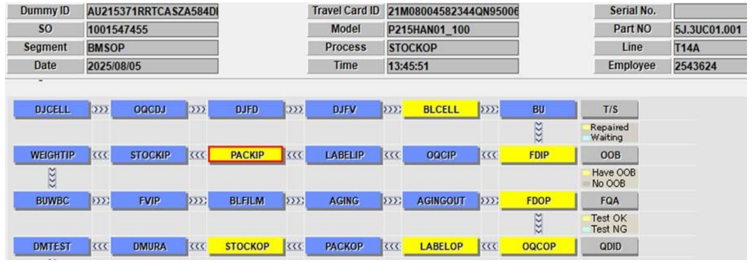
- Among the three defective samples, the panel below shows the most stable flicker reproduction:

S/N: Q55A25P5KLZZZZ1900

SHIPPING_NO	INVOICE_NO	CARTON_NO	PALLET_NO	MODEL_NO	FINAL_GRADE	PART_NO	STOCK_IN_WEEK
Q55A25P5KLZZZZ1900	X511072984	ZZ19002580500328	Z19W2580500080	P215HAN01.1	Z	97.21P03.100	202532
Q55A4736QLZZZZ1900	X511072984	ZZ19002580500735	Z19W2580500149	P215HAN01.1	Z	97.21P03.100	202532
Q55A40P8XHZZZZ1900	X511072984	ZZ19002580500179	Z19W2580500149	P215HAN01.1	Z	97.21P03.100	202532

Problem Statement

- History

<table>
<tr><th>No.</th><th>Panel SN</th><th>Process Flow</th><th>Process time</th><th>Operator</th></tr>
<tr><td>1</td><td>AU215371RRTCASZA5849E70</td><td>
</td><td>2025-8-5</td><td>2563175</td></tr>
<tr><td>2</td><td>AU215371RRTCASZA584B5V0</td><td>
</td><td>2025-8-5</td><td>2563175</td></tr>
<tr><td>3</td><td>AU215371RRTCASZA584DIK0</td><td>
</td><td>2025-8-5</td><td>2444054</td></tr>
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Root cause analysis

1) JUNGLIM found 3 pcs P215HAN01.1 panel with Flicker.
2) After AUO check how much the Vcom value fluctuates due to ESD charge on panels that exhibit flicker. We confirmed that panels with flicker show much greater Vcom fluctuations compared to normal panels, and that these fluctuations can reach the margin limits. This was identified as the cause of the observed flicker.
3) Root cause : Vcom was not adjusted to the very best value.

No	SN	Vcom readback	V line phenomenon	Re-adjusting	V line phenomenon
1	Q55A25P5KLZZZZ1900	73	Serious	67	None
2	Q55A4736QLZZZZ1900	75	Serious	70	None
3	Q55A40P8XHZZZZ1900	73	Serious	67	None

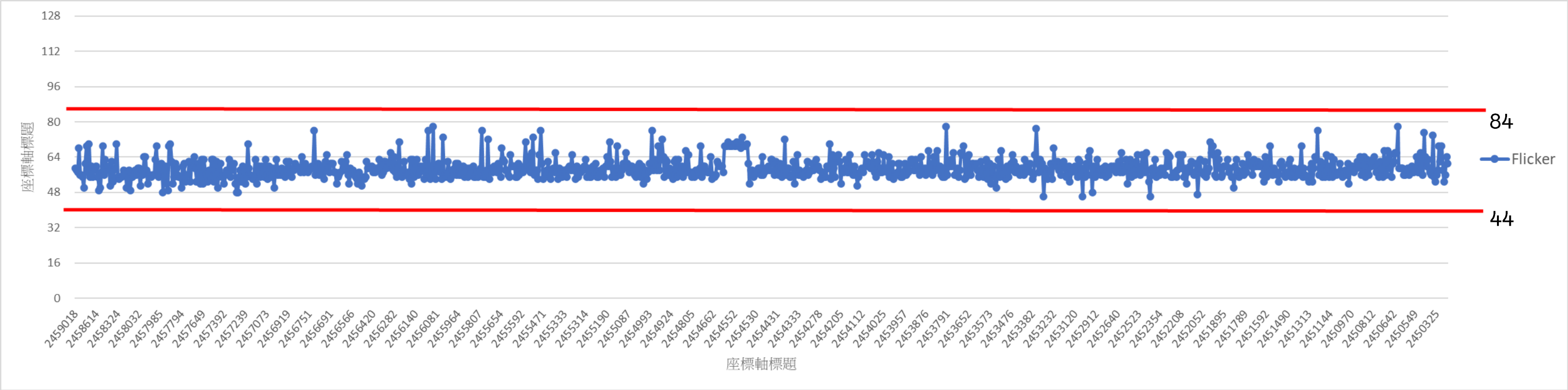
Therefore, we attempted to adjust the Vcom so that flicker would not occur, even if the Vcom fluctuates due to ESD charge.

Improvement action

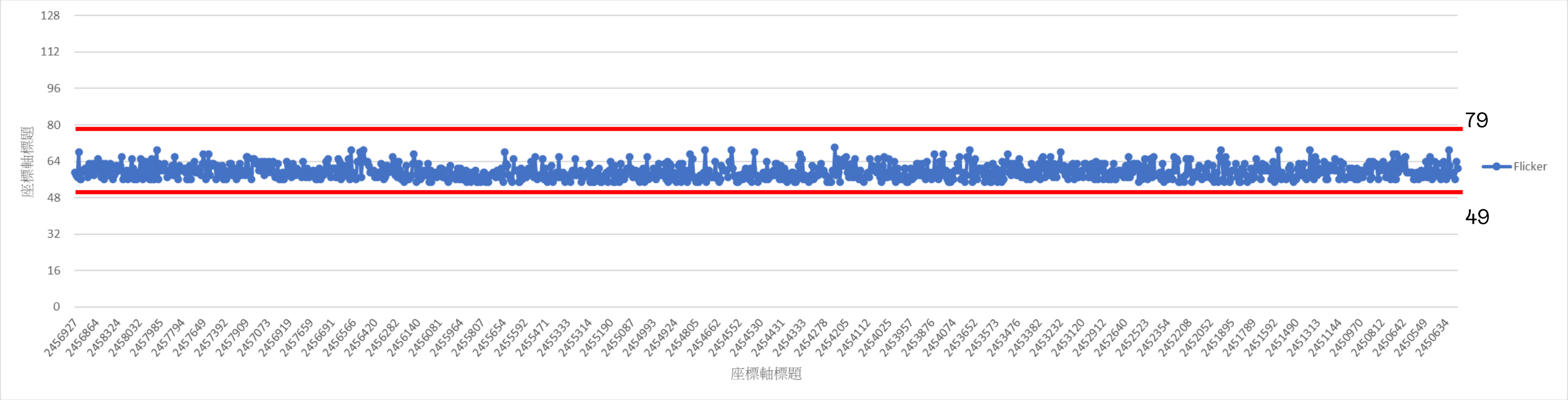
■ Action1: Adjust Flicker value:

Vcom values tightened from ±20 to ±15 for upper and lower limit control.

Before



After

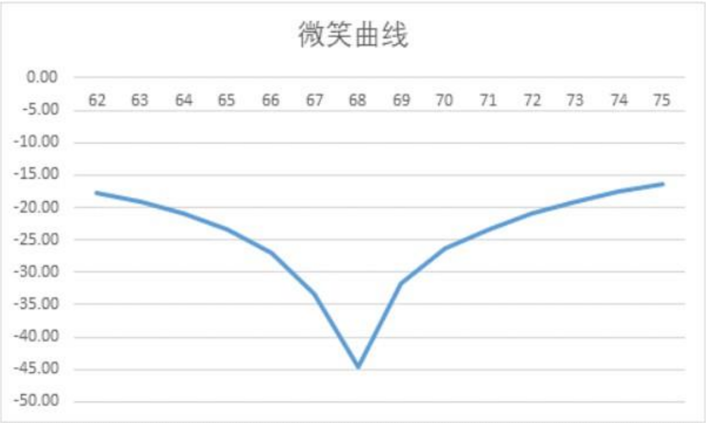


Verify Action

Action1: Adjust Flicker value:

After verification and multiple tests conducted by AUO, it has been confirmed that Vcom values of 66–70 are within the normal range. We plan to manage and ship products within this range to Junglim.

Vcom value	Q'ty	Sum
66~70	61	OK
Sum	**61**	



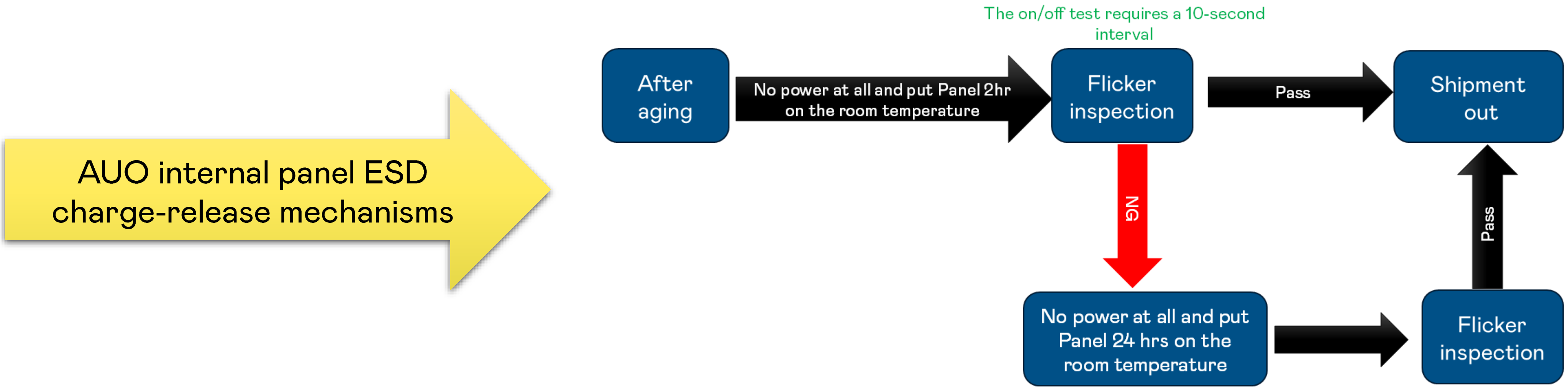
Vcom readback	Vcom	dB	V line
62	4.75	-17.85	Visible
63	4.77	-19.17	Visible
64	4.79	-20.98	Visible
65	4.81	-23.39	Minor
66	4.83	-27.1	None Visible
67	4.86	-33.4	None Visible
68	4.88	-44.6	None Visible
69	4.91	-31.68	None Visible
70	4.93	-26.3	None Visible
71	4.96	-23.45	Minor
72	4.98	-20.87	Visible
73	5.01	-19	Visible
74	5.03	-17.46	Visible
75	5.06	-16.3	Visible

Improvement action

■ <u>**Action2: Improve ESD release process to avoid flicker panel.**</u>

AUO improve Panel ESD Charge-Release Procedure on 2026/01/E.

And after tighten Vcom values, the yield loss is 12.9%. This indicates that the current mechanism can effectively prevent the outflow of flicker panels.



Tap Into The Possibilities